

Really touchy Touchscreens

The importance of being tactile

Nimish Chandiramani

readersletters@thinkdigit.com

We've loved our touchscreens for as long as we can remember.

After all, what can be more gratifying than poking at a screen and making something happen? And there's also multi-touch, of course. Whether it's Jeff Han's incredible demo or the iPhone's pinch gestures, there are many oohs to be emitted.

But in all our geek-gasms, we've ignored what touchscreens have really become – distractions. If you've got a touchscreen phone, for example, you've probably forgotten about the days when you could type out a message without looking at the phone (and so what if "cool" became "book"?). It's not such a big deal with cellphones, but with touchscreens coming to cars, the implications of having to look at a screen to interact with it become much more dangerous.

And so, Chris Harrison (CH) and Scott Hudson in the

Human Computer Interaction Institute at Carnegie Mellon University (CMU) came up with a design to bring the usefulness of physical buttons to the flexibility of touchscreens.

The concept is straightforward: use a latex membrane to make the touchscreen "puff up" in the shape of buttons. You get the ability to display anything on the touchscreen, but you still have big squishy buttons you can push. The idea itself doesn't beget as much awe as it does the question, "How did it take so long for someone to figure this out?"

But enough of that. We (d) spoke to Harrison recently, and so we'll let him do the talking.

d How did this project come about?

CH: The big dream in HCI [Ed. notes: Human-computer Interaction] is to have a tactile display. People have been looking at the problem for a very long time, and have come up with some really complicated solutions for this. There are three distinct parts to building a tactile display:

being able to render content, being able to do the tactile, and being able to accept input from the user. Most people are able to do two of the three. Microsoft Surface, for example, does input really well, but doesn't do tactile. If you look at interfaces in cars, the physical ones can do the tactile and the input, but they don't have the ability to display graphics.

To do all three is a really tough problem. And so we looked at what technologies were out there, and we came to this solution, which is a bit unusual in that it requires micro-bladders of air, but we were able to take advantage of two really good technologies: rear-projection, and multi-touch with infrared. This is very similar to the setup you see on a multi-touch table like Microsoft Surface, but we've added the extra tactile layer on top of it. So we get all the same great multi-touch capabilities, and it's rear-projected so we don't get the occlusion from hands that you would get from a front-projected screen.

It's also inexpensive; we were able to do this with materials in our labs

– primarily acrylics or plastic – and we were able to do sophisticated cut-outs using our laser cutter. It's somewhat specialised, but the material costs per display were no more than typical touchscreens.

d How is your solution better than existing tactile technologies?

CH: Well, unfortunately there aren't many existing tactile technologies. Among the ones that do exist, the most sophisticated tactile screens have a series of servo motors operating behind the screen, but any time you put actuators behind a screen, you run into a few problems. One is that if you're using rear projection, you're going to occlude the screen with lots of motors [Ed. notes: which means no multi-touch]. And if you're using an LCD screen, it's very difficult to have contiguous areas of space without breaking up the screen – having holes in the LCD where you can have motors. Also, from a scale point of view, if you have a 10x10 grid of little servo motors that cost \$5 (Rs 250) apiece, you're paying for 100 servos.



The researchers' ATM interface prototype demonstrates what they can do with the display